

ROHAN VENKATESH

US CITIZEN

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| **Portfolio** - rohan-v.com

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PROFESSIONAL SUMMARY

Aerospace engineering student with experience in propulsion, systems engineering, and computational research spanning CFD, MBSE, hydrogen systems, and defense technology analysis.

EDUCATION

Purdue University, West Lafayette, IN

Aug 2023 - May 2027

Bachelor of Science in Aeronautical and Astronautical Engineering

GPA: 3.76

Relevant Coursework - Propulsion, Fluid Dynamics, Thermodynamics, Aerodynamics, Aerospace Design, Dynamics, Structures, Controls, Statistics

TECHNICAL SKILLS

Programming & Analysis - Python, MATLAB, Simulink, C, MySQL

Engineering Software - ANSYS Fluent, Siemens NX, Capella, STK, OpenRocket

Research & Other Tools - Scopus, Ollama/LLMs, Microsoft Excel, xlwings

RELEVANT EXPERIENCE

Space Force Research Intern, Air Force Institute of Technology - Dayton, OH

May 2026 - Aug 2026

- Conducted Space Force-funded research within a 6-intern team, using computational and bibliometric methods to assess technological superiority and identify emerging aerospace and defense technologies.
- Developed MATLAB methods for Applied AI and Scaled Hypersonics, supporting comparison of technology growth, research impact, and emerging technical fields.

Undergraduate Researcher, Li Qiao Group - Purdue University

Aug 2025 - May 2026

- Worked within a 5-student team to develop an autonomous docking and hydrogen-refueling system for NASA-funded “Clean Forever-Flying Drones,” supporting long-duration hydrogen UAV operation.
- Converted 3g vertical, 1.5g lateral, and 25 m/s gust conditions into an approximately 3.6 kN docking design load and evaluated latching concepts through weighted trade studies.
- Compared concepts across load capacity, reliability, integration, hydrogen compatibility, and mass, supporting selection of a lightweight push-lock / ball-lock pin architecture.

MBSE Intern, BlueKei Solutions - Pune, India

Jul 2025 - Aug 2025

- Supported MBSE development of reusable launch vehicle landing gear through requirements, functional architecture, verification planning, and failure analysis.
- Developed an AI-enabled Excel/xlwings tool for requirement-quality analysis, traceable functions, V&V criteria, test recommendations, and FMEA outputs, reducing repetitive manual analysis.

Undergraduate Teaching Assistant, Purdue University

Aug 2024 - May 2025

- Led Physics 172 and 272 laboratory instruction for 60+ students across two lab sections, guiding application of mechanics, electromagnetism, and Python-based computational physics.

PERSONAL PROJECTS

Pressure-Fed LOX/CH₄ Rocket Engine Feed System

- Developed a Python model for a 1 kN pressure-fed LOX/CH₄ engine, sizing a 0.363 kg/s propellant flow system and predicting required tank pressures of approximately 26–27 bar.
- Performed parametric studies of feedline diameter, tank pressure, and injector pressure drop to evaluate design sensitivities and pressure margin.
- Built an approximately 75,000-element ANSYS Fluent LOX feedline model, verified close inlet/outlet mass-flow agreement, and compared CFD pressure behavior against the analytical model.

Rocketry - Tripoli Level 1 Certification

- Earned Tripoli Level 1 high-power rocketry certification at Thunderstruck 2024 using a LOC IV rocket and AeroTech H100W motor.
- Used OpenRocket to compare motor configurations and wind conditions and evaluate flight performance, stability, and recovery deployment.